Software Test Plan (STP)

Project: Demand Forecasting System (DFS)
Version: 1.0
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Methodology: Agile (Scrum, 1-week sprints)

Team members

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Testing runs inside every sprint and in the CI/CD pipeline.

Revision history

Version	Date	Author	Change summary	Approval
1.0	28-10-2026	DFS Team	Initial test plan with test-case catalogue	

Approvals

Role	Name	Signature / Email	Date
QA Lead			
Dev Lead			
Product Owner			

1. Introduction

Purpose: This document defines the test plan for the Demand Forecasting System (DFS) v1.0. It describes the objectives, scope, strategy, resources, schedule and responsibilities for testing.

Scope: Testing covers authentication and access control, master data (products, materials, BOM, suppliers), data import, inventory, demand forecasting (functional behaviour and forecast accuracy), raw-material requirement planning, purchase recommendations, alerts, dashboard and reports, performance and security. The internals of the hosting platform and of the external SMTP provider are excluded.

References: DFS SRS v1.0, DFS Software Architecture and Design Specification v1.0, OWASP Top 10, OWASP ASVS, WCAG 2.1.

Definitions: DFS (Demand Forecasting System), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), BOM (Bill of Materials), MRP (Material Requirements Planning), MAPE (Mean Absolute Percentage Error), RBAC (Role-Based Access Control), CI/CD (Continuous Integration / Continuous Delivery), UAT (User Acceptance Testing), TC (Test Case).

2. Test Items

- Authentication module (login, session timeout, password reset, roles, lockout)
- Master data module (products, raw materials, BOM, suppliers, material-supplier links)
- Data ingestion module (sales import, error report, market indicators)
- Inventory module (stock movements, safety stock, reorder point)
- Forecasting engine and worker (models, back-test, model selection, intervals, overrides, approval)
- Requirements (MRP) and procurement planner (BOM explosion, supplier split, approval, draft POs)
- Alert service and email notifications
- Reporting and export (dashboard KPIs, CSV/PDF)
- REST API layer, database and web user interface

3. Features to be Tested

Features are mapped to the SRS requirement IDs.

Req ID	Feature
DFS-F-001	Login and session timeout
DFS-F-002	Role-based access
DFS-F-003	Password reset
DFS-F-004	Products, materials and BOM
DFS-F-005	Supplier management
DFS-F-006	Material-supplier mapping
DFS-F-007	Import sales history
DFS-F-008	Inventory and stock movements
DFS-F-009	Market indicators
DFS-F-010	Generate forecast
DFS-F-011	Back-test and best-model selection
DFS-F-012	Forecast chart and intervals
DFS-F-013	Forecast override and approval
DFS-F-014	Short-history fallback
DFS-F-015	BOM explosion (MRP)
DFS-F-016	Safety stock and reorder point
DFS-F-017	Purchase recommendations
DFS-F-018	Approve recommendation / draft PO
DFS-F-019	Low-stock and accuracy alerts
DFS-F-020	Dashboard KPIs
DFS-F-021	CSV / PDF export
DFS-NF-001	Response time
DFS-NF-002	Forecast run time
DFS-NF-003	Forecast accuracy
DFS-NF-004	Availability and backup
DFS-NF-005	Responsive and accessible UI
DFS-NF-006	80% test coverage
DFS-NF-007	Data volume

Req ID	Feature
DFS-SR-001 to 006	Security requirements (see Section 5.1)

4. Features Not to be Tested

- Internal implementation of the external SMTP provider (only that DFS hands over the email correctly is tested)
- Hosting platform infrastructure and its uptime guarantee (availability is monitored, not stress-broken)
- Correctness of third-party libraries (statsmodels, pandas, chart and PDF libraries) beyond how DFS uses them
- The company's existing ERP system that produces the import files
- Features outside SRS v1.0 (EDI to suppliers, payments, production scheduling, deep-learning models)

5. Test Approach / Strategy

Levels:

- **Unit tests:** functions and services, e.g. safety-stock formula, BOM explosion, MOQ rounding, each forecasting model with fixed seeds
- **Integration tests:** API with database, API with Redis and the forecast worker, API with the fake SMTP server
- **System tests:** end-to-end flows in the browser (import -> forecast -> approve -> requirements -> purchase recommendation -> export)
- **Acceptance tests:** UAT against the SRS acceptance criteria with team members acting as each role

Types:

- Functional testing (all DFS-F requirements)
- **Forecast accuracy / model validation:** back-testing on the reference data set, comparison with a naive baseline, check that intervals contain the forecast (DFS-F-011, DFS-NF-003)
- **Calculation verification:** MRP, safety stock, reorder point and supplier split compared with an independently built spreadsheet
- **Data quality testing:** malformed, duplicate, missing and extreme values in import files
- Regression testing (automated suite run on every push)
- Performance and load testing (response time, forecast run time, data volume)
- Usability, compatibility and accessibility testing
- Security testing (Section 5.1)

Entry criteria: stable build available from the pipeline, reference data set and test accounts loaded, environment ready, sprint requirements baselined.

Exit criteria: all planned test cases executed; no critical or high-severity defects open; all High-priority requirements passed; mean MAPE on the reference data set <= 20% (DFS-NF-003); code coverage >= 80%; all SRS acceptance criteria satisfied.

5.1 Security Validation

Req ID	What is validated	Method
DFS-SR-001	TLS 1.2+	testssl.sh scan; open site over HTTP
DFS-SR-002	Password hashing and lockout	Database inspection, scripted login attempts
DFS-SR-003	Server-side authorisation	API tests with tokens of each role
DFS-SR-004	Input and file validation	Manual payloads, malicious files, OWASP ZAP scan

Req ID	What is validated	Method
DFS-SR-005	Audit log	Perform audited actions and inspect log; attempt update via API
DFS-SR-006	Rate limiting	Scripted repeated requests

Security objectives verified: SO-1 Confidentiality (SR-001, 002, 003, 006), SO-2 Integrity (SR-003, 004) and SO-3 Accountability (SR-005). Additionally, Bandit (Python static analysis) and dependency vulnerability checks run in the pipeline from Sprint 1.

6. Test Environment

Hardware: developer laptops, cloud CI runners and a cloud-hosted staging VM (2 vCPU, 4 GB RAM) matching production.

Software: DFS front end and REST API, Celery worker, PostgreSQL 16 and Redis 7 test instances (Docker Compose), Chrome, Edge and Firefox, tablet emulation.

Tools:

Purpose	Tool
Back-end unit testing and coverage	pytest, pytest-cov
Front-end unit testing	Vitest, React Testing Library
API testing	Postman / Newman
UI / end-to-end automation	Playwright
Performance and load testing	Locust
Security scanning	OWASP ZAP, testssl.sh, Bandit
Accessibility	Lighthouse, axe DevTools
Code quality	SonarCloud
Email testing	Mailpit (fake SMTP server)
CI/CD	GitHub Actions
Defect tracking and backlog	GitHub Issues / Projects

Test data:

- One test account per role (Administrator, Demand Planner, Inventory Manager, Procurement Manager).
- **Reference data set:** 50 products, 30 raw materials, 10 suppliers and 36 months of sales generated with a known trend, seasonality and noise (fixed random seed), together with a spreadsheet of hand-calculated expected requirements, safety stock, reorder points and purchase quantities.
- Edge-case files: empty file, wrong columns, unknown product codes, negative quantities, duplicate periods, 20 MB file, renamed executable, macro-enabled workbook.
- Short-history products (6 months) and products with zero demand in some periods.
- Volume data set: 1 million sales rows and 2,000 products for DFS-NF-007.

7. Test Schedule

Testing is repeated in every sprint (1-week sprints).

Milestone	Sprint
Test plan, CI test skeleton, reference data set generator	Sprint 0

Milestone	Sprint
Test design and execution for Auth and Master Data	Sprint 1
Test design and execution for Data Import and Inventory	Sprint 2
Forecasting tests and model-accuracy validation	Sprint 3
MRP, procurement and alert tests	Sprint 4
Reporting, performance, security and accessibility test execution	Sprint 5
Full regression run and defect closure	Sprint 5
UAT	Before final submission

8. Test Deliverables

- Test Plan (this document)
- Test cases (manual and automated)
- Automated test scripts (unit, API, end-to-end, performance)
- Reference data set and expected-results spreadsheet
- Model validation (back-test) report
- Test execution logs and CI pipeline reports
- Defect reports
- Test Summary Report

9. Roles and Responsibilities

Roles rotate per sprint; the table shows the initial assignment.

Role	Name	Responsibility
QA Lead	G Ganikhasri	Prepare plan, coordinate execution, report status
Test Engineer	Gagana C	Design and execute test cases, maintain reference data, log defects
Developer	Dhyeya Aneesh	Write unit tests, support defect fixes and triage
Product Owner	Bandugula Sai Rithesh Reddy	Approve test results and sign off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds; test each module as soon as it is merged
Forecast results hard to verify (no single 'correct' answer)	Use the synthetic reference data set with known patterns, fixed seeds and accuracy thresholds instead of exact values
Lack of realistic company data	Generate synthetic data; review its patterns with the instructor
Test environment downtime	Run the stack locally with Docker Compose as a backup
Long forecast runs slow down the pipeline	Use small data sets in CI; run full-volume tests on staging in a scheduled window
Flaky UI tests	Stable selectors, retries, keep end-to-end tests focused on key flows
Insufficient time for security testing	Add ZAP and Bandit scans to the pipeline from Sprint 1

11. Assumptions and Dependencies

- The CI pipeline and a staging environment are available from Sprint 0
- A fake SMTP server (Mailpit) is available for email tests
- The reference data set and expected-results spreadsheet are ready before Sprint 3
- Requirements stay stable within a sprint

12. Suspension and Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours
- The build is so unstable that more than 30% of test cases are blocked
- A critical defect blocks login, data import or forecast runs

Resume testing when:

- Blocking defects are fixed and verified
- The environment is stable and a new build passes the smoke tests

13. Test Case Management and Traceability

The RTM maps every SRS requirement to one or more test cases so that no requirement is left untested.

13.1 Test Case Catalogue

Test case ID	Req ID	Description	Expected result	Type
TC-Auth-01	DFS-F-001	Log in with valid and invalid credentials; stay idle for 30 min	Valid login opens dashboard; invalid shows generic error; idle session expires	Functional
TC-Auth-02	DFS-F-002	Log in as each role and open pages/APIs of other roles	Access denied (403) for pages outside the role	Functional
TC-Auth-03	DFS-F-003	Request reset, use link, reuse link, use link after 30 min	Link works once; expired/reused link rejected	Functional
TC-MD-01	DFS-F-004	Create product with BOM; try duplicate code and BOM qty 0	Valid item saved; duplicate code and qty <= 0 rejected	Functional
TC-MD-02	DFS-F-005	Add and edit a supplier with valid and invalid lead time, MOQ and price	Valid supplier saved; invalid values rejected with field errors	Functional
TC-MD-03	DFS-F-006	Link a material to 3 suppliers, set preferred, deactivate all suppliers	Mapping saved; one preferred; material without supplier flagged	Functional
TC-Data-01	DFS-F-007	Import a file with 100 valid and 5 invalid rows; re-import same file	100 rows stored, 5 listed in error report; re-import causes no duplicates	Functional
TC-Data-02	DFS-F-008	Post receipts, issues and an adjustment; post an issue larger than stock	On-hand matches hand calculation; over-issue rejected	Functional
TC-Data-03	DFS-F-009	Enter a promotion and price change for a product and run a forecast	Indicators saved and listed as inputs of the forecast run	Functional
TC-Fc-01	DFS-F-010	Run a 6-month forecast for 10 products	Run reaches DONE; 6 forecast values per product	Functional
TC-Fc-02	DFS-F-011	Run forecast on the reference data set and compare chosen model and MAPE with expected values	Selected model and metrics match the expected results	Functional
TC-Fc-03	DFS-F-012	Open the forecast page for a product	Chart matches table; intervals contain the forecast	Functional
TC-Fc-04	DFS-F-013	Override a value with and without a reason, approve, try to edit again	No-reason override rejected; approved version locked; audit entry created	Functional
TC-Fc-05	DFS-F-014	Run forecast for a product with 6 months of history	Moving-average forecast produced and flagged Low confidence	Functional

Test case ID	Req ID	Description	Expected result	Type
TC-Proc-01	DFS-F-015	Approve a forecast and compute requirements for the reference data set	Gross and net requirements match hand calculation	Functional
TC-Inv-01	DFS-F-016	Compute safety stock and ROP for 3 materials at 90% and 95% service level	Values match the spreadsheet calculation	Functional
TC-Proc-02	DFS-F-017	Generate recommendations where demand exceeds the preferred supplier's capacity	Quantity split across suppliers; MOQ and capacity respected; dates correct	Functional
TC-Proc-03	DFS-F-018	Approve one recommendation, reject one without and with comment, re-approve	Draft POs created; reject needs comment; second approval blocked (409)	Functional
TC-Inv-02	DFS-F-019	Post an issue that takes a material below its reorder point	One dashboard alert and one email within 5 min	Functional
TC-Rep-01	DFS-F-020	Open the dashboard with the reference data set	All KPI values match the reference figures	Functional
TC-Rep-02	DFS-F-021	Export each report as CSV and PDF; try export with a role without access	Files complete and correct; unauthorised export denied	Functional
TC-Perf-01	DFS-NF-001	Load test with 50 virtual users for 15 minutes	95th percentile response <= 2 s, error rate < 1%	Performance
TC-Perf-02	DFS-NF-002	Start a 500-product forecast run while 50 users browse	Run finishes <= 5 min; browsing stays within 2 s	Performance
TC-Fc-06	DFS-NF-003	Run back-test on the reference data set and compare with naive baseline	Mean MAPE <= 20% and better than naive	Forecast quality
TC-Rel-01	DFS-NF-004	Review monthly uptime report and perform a backup-restore drill	Availability >= 99.5%; restore within 4 h with <= 24 h data loss	Reliability
TC-UX-01	DFS-NF-005	Check main pages at 768, 1280 and 1920 px on 3 browsers; run Lighthouse	No broken layout; accessibility score >= 90	Usability
CI coverage report	DFS-NF-006	Run the coverage check in the pipeline	Build fails when coverage is below 80%	Maintainability
TC-Perf-03	DFS-NF-007	Seed 1 M sales rows and 2,000 products, repeat TC-Perf-01	95th percentile still <= 2 s	Scalability
TC-Sec-01	DFS-SR-001	Scan the TLS configuration and open the site over HTTP	HTTP redirects to HTTPS; TLS < 1.2 refused	Security
TC-Sec-02	DFS-SR-002	Inspect stored passwords; try 6 wrong passwords	Only hashes stored; account locked after 5 failures	Security
TC-Sec-03	DFS-SR-003	Call approval, override and admin APIs with tokens of other roles	403 returned; no data changed or exposed	Security
TC-Sec-04	DFS-SR-004	Upload .exe renamed to .csv, 20 MB file, macro .xlsm; enter SQL/script payloads; export a cell starting with '='	Rejected or escaped; exported formula shown as text	Security
TC-Sec-05	DFS-SR-005	Perform each audited action and inspect the audit log	One entry per action with user, time, old and new value; log cannot be edited	Security
TC-Sec-06	DFS-SR-006	Send 20 login requests in 1 minute and 150 API calls in 1 minute	HTTP 429 after the limit	Security

13.2 Traceability Summary

Every requirement ID in Section 3 has at least one test case in the catalogue above (21 functional, 7 non-functional and 6 security requirements; 34 test cases). The full requirement-to-module-to-test mapping is kept in the RTM of the SRS (Section 8), where the status column is updated as tests pass.

14. Test Metrics and Reporting

Metrics collected:

- Percentage of test cases executed, passed and failed

- Defect density and defect aging
- Requirement coverage (requirements with at least one passed test)
- Code coverage from the CI pipeline
- Forecast accuracy on the reference data set (mean MAPE per model and for the selected model)
- Pipeline success rate

Reports:

- Daily execution status during test sprints
- Model validation report
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner		